SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Extruded Polystyrene (XPS) Foam Board, target R-10.0 (skipped: target R-value already satisfied) **Roof upgrade:**

		• Roof insulation: Blown Mineral Wool, target R-15.0 **Window upgrade:** • 1-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Window frame replacement: wood-aluminium window frame **Door upgrade:** • Door replacement: glass door
Scenario 2	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Expanded Polystyrene (EPS) Foam Board, target R-30.0 **Roof upgrade:** • Roof insulation: Polyiso Insulation Foam Board, target R-25.0 **Window upgrade:** • 2-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: polyurethane • Window frame replacement: wood window frame **Door upgrade:** • Door replacement: polystyrene core steel door • Bottom seal: brush weatherstrip • Top/side seal: silicone adhesive smoke gasket
Scenario 3	wall + roof + window + door	**Wall upgrade:**

- Exterior wall insulation: Fiberglass Batts, target R-20.4

Roof upgrade:

- Roof insulation: Blown Fiberglass, target R-34.5

Window upgrade:

- 3-pane replacement (skipped: SimpleGlazing windows)
- Weatherstrip application (skipped: no OperableWindow)
- Glazing film application (skipped: SimpleGlazing windows)
- Caulking application: acrylic
- Window frame replacement: wood window frame

Door upgrade:

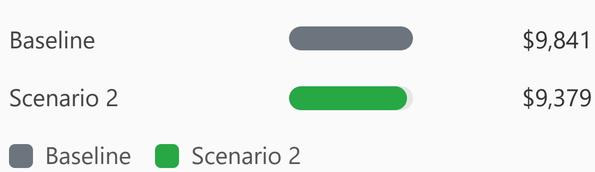
- Door replacement: wooden door
- Bottom seal: automatic door bottom
- Top/side seal: jamb weatherstrip

Annual Energy Analysis

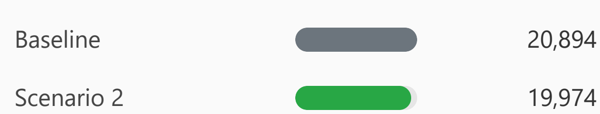
Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	228.19	**8.35**	**3.53%**
Scenario 2	Total Site Energy (GJ)	236.54	222.75	**13.79**	**5.83%**
Scenario 3	Total Site Energy (GJ)	236.54	223.47	**13.07**	**5.53%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr) Comparison



Max saving: **$-462 (-4.69%)**

Baseline Scenario 2

Max saving: **-920 kg CO_2e (-4.40%)**

Min Retrofit Construction Cost

$60,016

Scenario 3

Min Retrofit Embodied Carbon

21,368 kg CO_2e

Embodied carbon intensity (ECI): 42 kgCO_2e/m^2

Scenario 3

Lowest Retrofit Cost Payback Period

139 years

Scenario 3

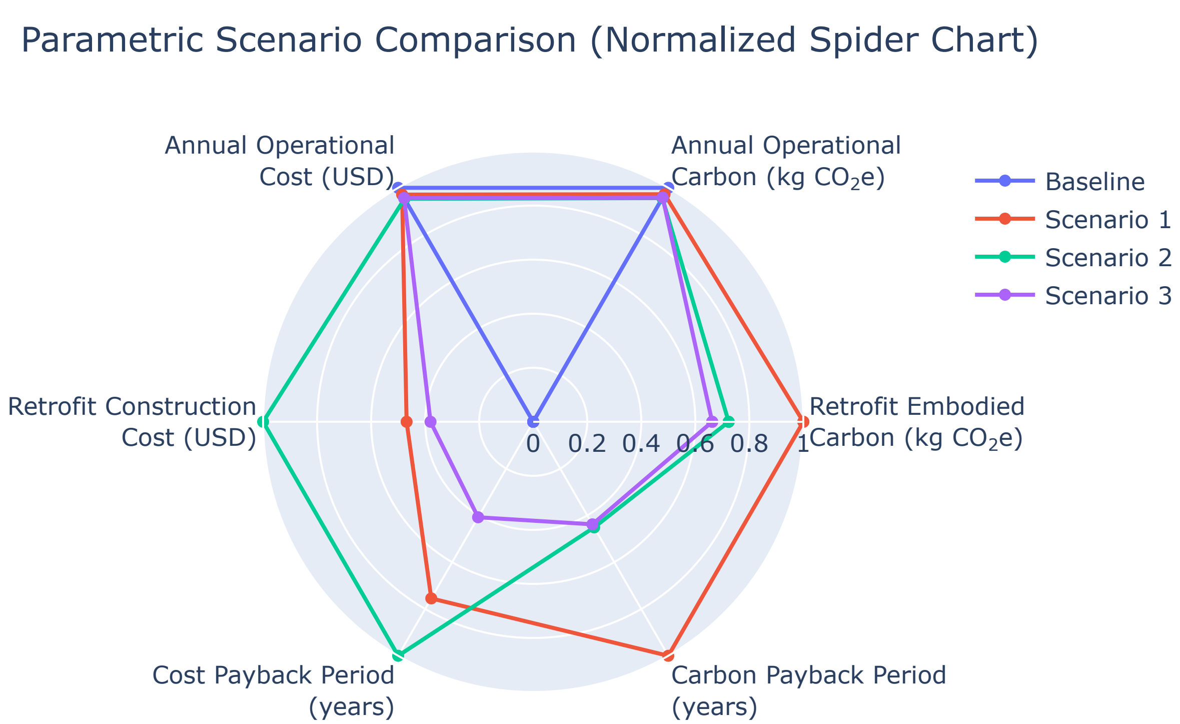
Lowest Retrofit Carbon Payback Period

25 years

Scenario 3

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 258 yrs

Scenario 2 342 yrs

Scenario 3 139 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr)).

Scenario 1 56 yrs

Scenario 2 25 yrs

Scenario 3 25 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Extruded Polystyrene (XPS) Foam Board	N/A	N/A	N/A	N/A	0.03	35	75
Scenario 1	Roof insulation	Blown Mineral Wool	61	599	0.10	N/A	0.03	90	30
Scenario 1	Window frame	wood-aluminium window frame	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	N/A	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	glass door	N/A	3.91	N/A	N/A	0.10	2,500	30
Scenario 2	Wall insulation	Expanded Polystyrene (EPS) Foam Board	21	282	N/A	N/A	0.03	20	60

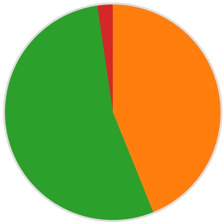
Scenario 2	Roof insulation	Polyiso Insulation Foam Board	104	599	0.17	N/A	0.03	29	60
Scenario 2	Window frame	wood window frame	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 2	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 2	Door	polystyrene core steel door	N/A	3.90	N/A	N/A	0.10	472	30
Scenario 2	Door bottom sealing	brush weatherstrip	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 2	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 3	Wall insulation	Fiberglass Batts	4.86	282	N/A	N/A	0.03	30	60
Scenario 3	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	68
Scenario 3	Window frame	wood window frame	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 3	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 3	Door	wooden door	N/A	3.90	N/A	N/A	0.15	638	30
Scenario 3	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 3	Door top/side sealing	jamb weatherstrip	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

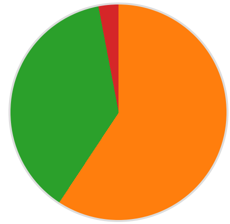
Cost breakdown by retrofit measure



- Roof: $32,470 (43.9%)
- Window: $39,802 (53.8%)
- Door: $1,717 (2.3%)

Total: $73,988

Carbon breakdown by retrofit measure



- Roof: 19,122 (59.3%)
- Window: 12,160 (37.7%)
- Door: 972 (3.0%)

Total: 32,255 kg CO_2e

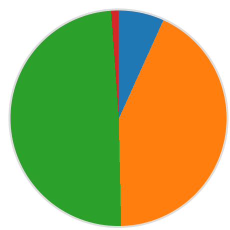
Scenario 2

Cost breakdown by retrofit measure

- Wall: $14,027 (8.9%)
- Roof: $93,306 (59.1%)
- Window: $39,346 (24.9%)
- Door: $11,163 (7.1%)

Total: $157,842

Carbon breakdown by retrofit measure

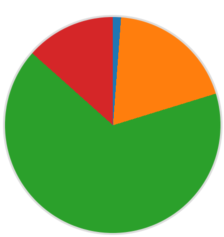


- Wall: 1,577 (6.8%)
- Roof: 10,008 (42.9%)
- Window: 11,489 (49.2%)
- Door: 265 (1.1%)

Total: 23,339 kg CO_2e

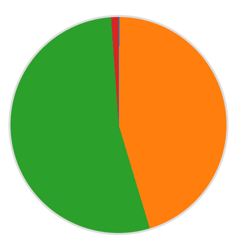
Scenario 3

Cost breakdown by retrofit measure



- Wall: $719 (1.2%)
- Roof: $11,441 (19.1%)
- Window: $39,802 (66.3%)

Carbon breakdown by retrofit measure



- Wall: 21 (0.1%)
- Roof: 9,669 (45.2%)
- Window: 11,437 (53.5%)

Door: $8,055 (13.4%)

Total: $60,016

Door: 241 (1.1%)

Total: 21,368 kg CO_2e

Scenario	Retrofit Embodied Carbon (kg CO_2e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO_2e)	Annual Operational Cost (USD)
Baseline	0.00	$0.00	20,894	$9,841
Scenario 1	32,255	$73,988	20,320	$9,554
Scenario 2	23,339	$157,842	19,974	$9,379
Scenario 3	21,368	$60,016	20,026	$9,411

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